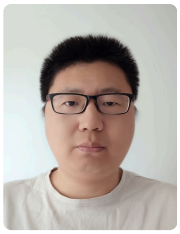
Heng Zhang

Target Role: Senior C/C++ Software Engineer, Embedded Software Engineer

Experience : 16 years
Date of Birth : 1987.12
Location : Qingdao, China
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Education

Dalian University of Technology | Software Engineering (Embedded Systems) | B.S. | 2006.09 – 2010.06 | CET-6 (557)

Summary

- 16 years in telecom software: 11 years Linux server networking, 5 years ARM/VxWorks RTOS protocol stacks
- Strong C/C++ on Linux, networking, and multithreading; system design, high concurrency, and performance tuning
- Docker/Kubernetes cloud-native migration and hitless upgrades for carrier-grade products
- Production troubleshooting for North American and European operators; rapid root-cause analysis on complex incidents
- Cross-team technical lead on multiple feature deliveries; fluent professional English
- Hands-on LLM inference and AI agent development

Technical Skills

Languages	C/C++ (C++11 at work, C++20 in personal projects), Python, Shell; familiar with Go
Networking	TCP/IP, SCTP, HTTP/2, Diameter; 3GPP 4G/5G core specifications
Systems	Linux multithreading and high-concurrency networking; perf, Valgrind, Wireshark
Cloud	Docker, Kubernetes, OpenShift, Helm; hitless rolling upgrades
Embedded	ARM Cortex-M; VxWorks RTOS embedded network stacks
AI	RAG, function calling, ReAct agents; Transformer inference, KV cache, quantization basics

Work Experience

Nokia (China) Qingdao · Senior Software Engineer — 2015.06 – Present

Modules	Owner of three core modules—4G/5G networking, DB cache, and common middleware; design reviews and quality gate
Leadership	Led 5–6 engineer squads on new features end to end; onboarding and internal tech talks
Cloud Native	Led container migration of core user-database C++ services; hitless upgrade design (2020 – Present)
Production	Field support for NA/EU operators; led T-Mobile major incident root-cause analysis (2017 – 2020)
Core Network	4G subscriber DB and IMS features in C/C++ per 3GPP (2015 – 2017)

Hitachi Information (Dalian) Co., Ltd. · Embedded Software Engineer — 2009.09 – 2015.05

Protocol Stacks	HTTP, UPnP/WSD, CIFS, SNMP on ARM/VxWorks per RFCs in C/C++
Print Protocols	Led AirPrint and Google Cloud Print; products passed AirPrint 1.1–1.5 certification
IPv6	Built IPv6 Ready Logo test lab; stack passed Phase-1/Phase-2
Internship	Embedded printer web UI (HTML/JS/CSS) and Selenium automation (2009 – 2010)

Project Experience

T-Mobile Core Signaling Latency Incident · Root-Cause Lead — 2019.06 – 2019.09

Background Fiber cut caused rerouting; Diameter/SCTP RTT exceeded 140 ms; ~2.29M LTE subscribers affected; VoLTE call volume down 25%

Contributions

- Mitigation: log/packet analysis with customer, isolated high-latency paths, traffic shift for fast recovery
- App layer: Wireshark compared healthy vs failing SCTP; send-side backlog and EAGAIN; single-threaded blocking send starved peers on same thread
- Root cause: lab replay at same RTT; controlled tests on NIC queues, socket buffers, bare metal vs VM; CentOS virtio multi-queue lacked true RSS → SCTP reordering and fast retransmit; CWND slow recovery at high RTT
- Fix: connection-level auto-isolation patch, buffer/timeout tuning; roadmap for single-queue, OS upgrade, and connection spreading

Results Service restored within 12 h; patch accepted in 72 h; ~10× single-connection throughput on single-queue setup; drove WAN latency lab and link-level monitoring

Core User-DB C++ Services on Kubernetes · Lead Developer — 2020.06 – Present

Background Multi-process Linux C++ user-database plane moved from VM/VNF to Kubernetes/OpenShift with hitless upgrades

Contributions

- Refactor: cloud-native rework of three C++ services—single-container multi-process, ConfigMap injection, hot reload
- Graceful shutdown: SIGTERM handling, Diameter/HTTP/2 drain, readiness/liveness probes for rolling upgrades
- Compatibility: mixed old/new Pod versions during rollout
- Release: Helm charts; Go CRD-based K8s operator; full-product rolling upgrade design

Results 99.9% success rate during upgrades; 2,500 QPS per Pod; production at T-Mobile, Verizon, Vodafone

RAG-Based Support & Ticket Q&A · Core Developer — 2025.10 – 2026.03

Background Manual ticket handling; duplicate issues; docs and past fixes hard to reuse

Contributions

- Ingestion pipeline: chunking, embedding, Chroma-backed retrieval; multi-format docs and site crawl
- RAG chain: query rewrite, retrieval, rerank/reject, cited generation
- Integration: SSE streaming; Jira REST sync and reconciliation

Results Resolved tickets as case cards; unified “docs + tickets” search; self-service for customers, faster first response internally

AI Ops Terminal · Solo Developer — 2026.06 – 2026.09

Background Ops work jumps across SSH sessions; generic AI lacks terminal context; unsafe autonomous commands

Contributions

- Electron + TypeScript multi-tab SSH with context-aware AI sidebar (F12 AI mode)
- ReAct agent with 30+ tools (shell, files, Git, planning); Plan/Agent/Execute modes
- Three-tier approval: read-only allowlist + dangerous-command detection; loop guard and step budget
- Context engineering: tool-scoped prompts, prefix-cache friendly; compression; sub-agents for multi-host read-only probes

Results Nokia internal AI innovation contest—2nd place; adopted inside the company

GPT-2 CPU Inference Engine (C++) · Personal Project — 2026.04 – Present

Background From-scratch Transformer inference on laptop CPU to learn LLM internals

Contributions

- Tensor ops with AVX2 SIMD (attention, GELU, LayerNorm, RMSNorm)
- Compute graph: topo execution, shape inference, buffer reuse, timing
- GPT-2 124M load, per-head KV cache, prefill/decode; manual forward + graph paths
- Sampling: temperature, Top-K/Top-P, presence/frequency penalties

Results ~111 tokens/s decode on Intel Core Ultra 5 135U (14 threads)

Apple AirPrint Development & Certification (ARM/VxWorks) · Protocol Engineer 2013.11 – 2015.05

Background Driverless printing for iPhone/iPad/macOS with Apple certification on resource-constrained ARM/VxWorks

Contributions

- Discovery: Bonjour (mDNS/DNS-SD) print service advertisement
- Jobs: IPP submit/status/error handling; PDF, JPEG, TIFF, URF pipelines
- RTOS integration: C/C++ on VxWorks alongside stack and drivers
- Certification: test harness; functional, interop, regression; AirPrint 1.1–1.5 deltas

Results Multiple printers certified 1.1–1.5; wireless print shipped on product lines

Patents, Honors & Certifications

Patent First inventor, PCT WO2025175539A1 (AKMA Authentication Based on Device Information)

Award Oct 2021 Nokia equity grant (1,000 shares) for outstanding performance

China Software Designer (Intermediate), National Computer Technology Examination

Red Hat Certified OpenShift Application Developer (240-240-408)